

Assignment – 3

1. ls -A (shows hidden file)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -A
12:05 . .bash_logout .config Downloads four .local myNewFile 'prac folder' sdm .ssh .sudo_as_admin_successful sunbeam_submission two
44.txt .bashrc Desktop file10.htm file4.htm file7.htm .gnupg Music one .profile Shivan snap sunbeam Templates three Videos
.bash_history .cache Documents five .lessht myName Pictures Public snap sunbeam
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

2. rename all *.txt to *.htm

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ rename 's/\.txt/\.htm/' *.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -a
. .bash_history .config .fhide file3.htm file7.htm four Music Pictures sdm .sudo_as_admin_successful three
12:05 .bash_logout Desktop file10.htm file4.htm file8.htm .gnupg myName 'prac folder' Shivan sunbeam
44.htm .bashrc Documents file1.htm files.htm file9.htm .lessht myNewFile .profile snap sunbeam_submission two
.cache Downloads file2.htm file6.htm five .local one Public snap sunbeam Templates
```

3. ln -s create symbolic link (shortcut) for a file

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ln -s /home/44.htm 44.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```



44.htm



44.txt

4. ln (Create a hard link for a file)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ln file3.htm file3
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```



file3



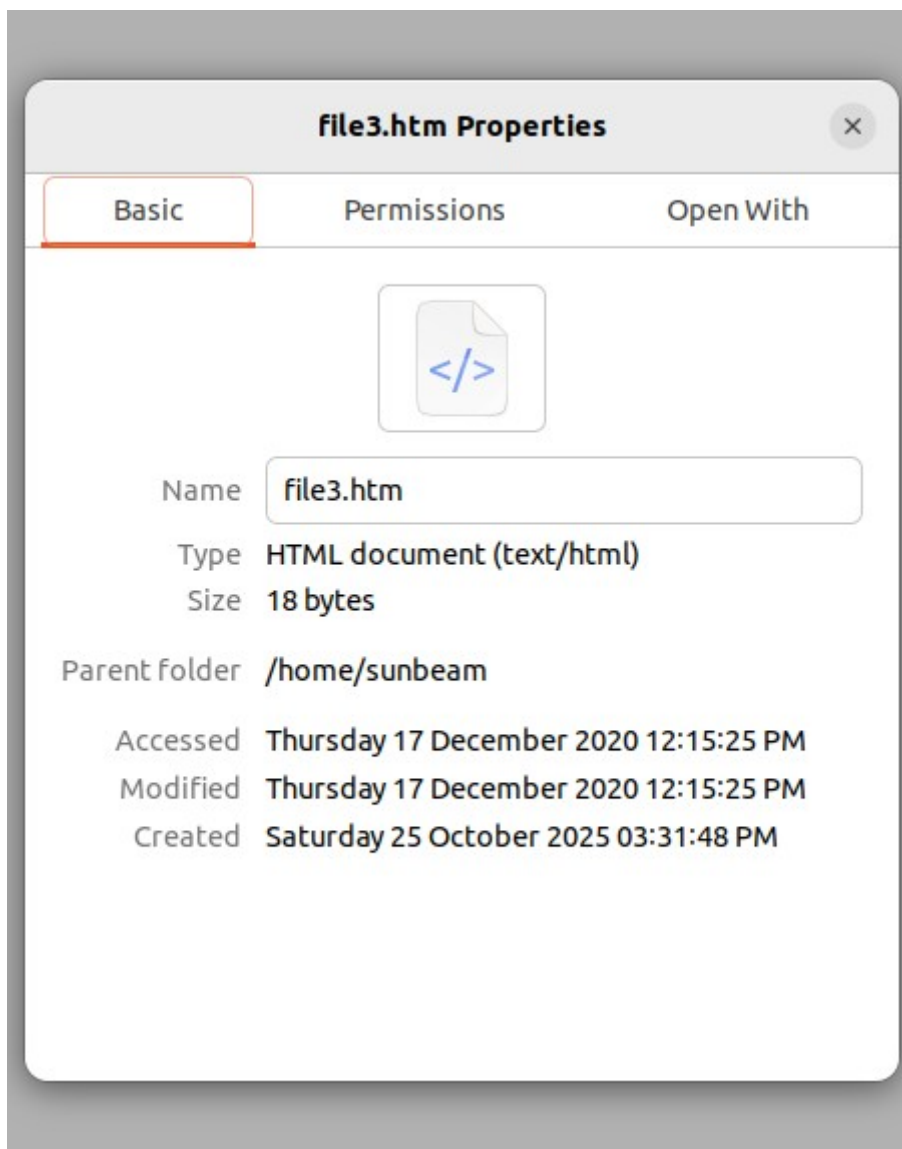
file3.htm

5. stat (Display the time Stamp of a file)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ stat file3.htm
File: file3.htm
Size: 18          Blocks: 8          IO Block: 4096   regular file
Device: 805h/2053d Inode: 1841212    Links: 2
Access: (0664/-rw-rw-r--)  Uid: ( 1000/ sunbeam)   Gid: ( 1000/ sunbeam)
Access: 2025-10-25 16:30:39.012844033 +0530
Modify: 2025-10-25 16:30:38.713888811 +0530
Change: 2025-10-25 16:30:38.713888811 +0530
 Birth: 2025-10-25 15:31:48.995828109 +0530
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

6. touch (change the time stamp of a file)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ man touch
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ touch -t 202012171215.25 file3.htm
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```



7. touch (Create empty files)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ touch file{11..20}.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls
12:05 Desktop file10.htm file13.txt file16.txt file19.txt file2.htm file4.htm file7.htm five myName Pictures sdm sunbeam three
44.htm Documents file11.txt file14.txt file17.txt file1.htm file3 file5.htm file8.htm four myNewfile 'prac folder' Shivam sunbeam_submission two
44.txt Downloads file12.txt file15.txt file18.txt file20.txt file3.htm file6.htm file9.htm Music one Public snap Templates Templates Videos
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

8. tail (Print all lines in a file from fifth line onwards)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ tail -15 file11.txt
this is sixth line
this is seventh line
this is eighth line
this is ninthe line
this is tenth line
this is eleventh line
this is twelfth line
this is thirteenth linejj
this is fourteen line
this is fifteen line
this is sixteen line
this is seventeenth line
this is eighteenth line
this is ninteenth line
this is twentieth line
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

9. init 0, poweroff (Shutdown (Only for super-user))

10. init 6, reboot (Reboot (Only for super-user))

12. whoami (find the current user)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ whoami
sunbeam
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

13. tty (find current terminal)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ tty
/dev/pts/0
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

14. who am i (Find current user and terminal)

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ who
sunbeam  tty2          2025-10-25 15:00 (tty2)
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

15. users (Find all users logged into the system).

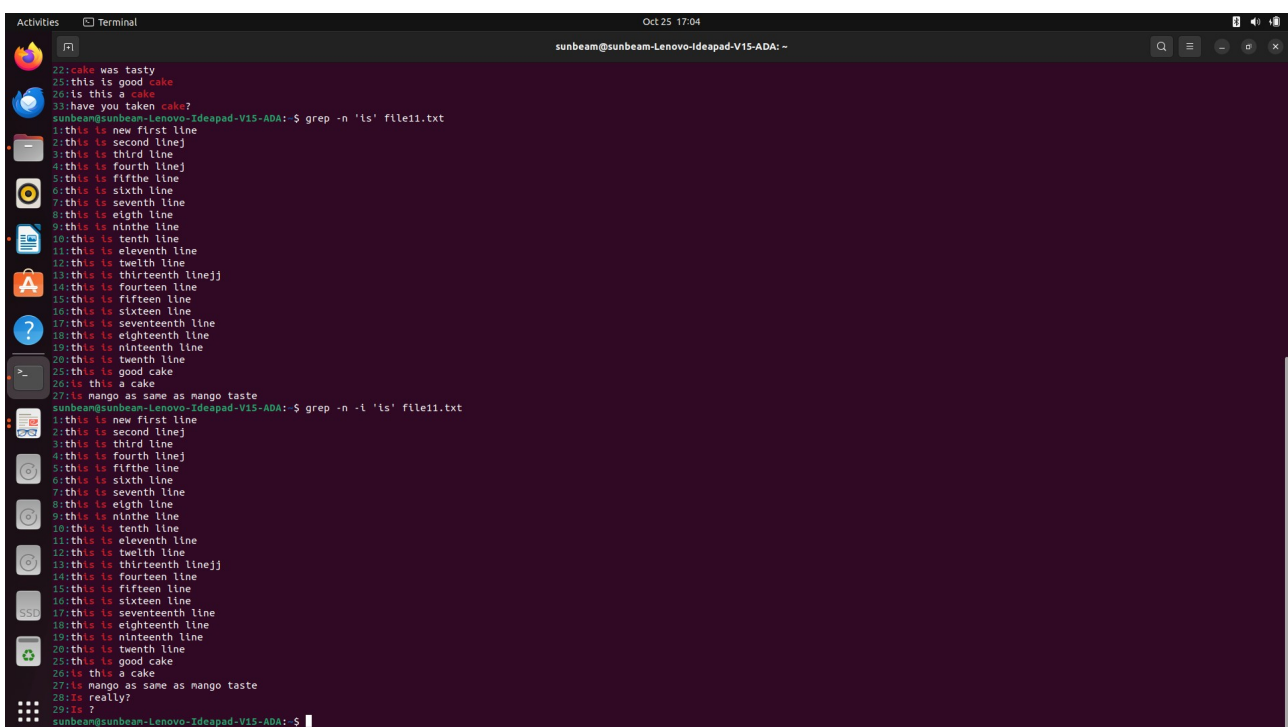
```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ who
sunbeam  tty2          2025-10-25 15:00 (tty2)
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

16. who (Find all users logged in and terminals in use).

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ who
sunbeam  tty2          2025-10-25 15:00 (tty2)
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

17. grep, egrep, fgrep (Find a given word/pattern from the file).

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ grep 'this' file11.txt
this is new first line
this is second linej
this is third line
this is fourth linej
this is fifthe line
this is sixth line
this is seventh line
this is eighth line
this is ninthe line
this is tenth line
this is eleventh line
this is twelfth line
this is thirteenth linejj
this is fourteen line
this is fifteen line
this is sixteen line
this is seventeenth line
this is eighteenth line
this is ninteenth line
this is twentieth line
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```



```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ grep -n 'is' file11.txt
1: this is new first line
2: this is second linej
3: this is third line
4: this is fourth linej
5: this is fifthe line
6: this is sixth line
7: this is seventh line
8: this is eighth line
9: this is ninthe line
10: this is tenth line
11: this is eleventh line
12: this is twelfth line
13: this is thirteenth linejj
14: this is fourteen line
15: this is fifteen line
16: this is sixteen line
17: this is seventeenth line
18: this is eighteenth line
19: this is ninteenth line
20: this is twentieth line
21: this is good cake
22: this is a cake
23: is mango as same as mango taste
24: is really?
25: is ?
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

```

sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ grep 'cake' file11.txt
taste of cake
cake was tasty
this is good cake
is this a cake
have you taken cake?
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ grep -n 'cake' file11.txt
21:taste of cake
22:cake was tasty
25:this is good cake
26:is this a cake
33:have you taken cake?
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ grep -n 'is' file11.txt

```

```

sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ egrep 't??s' file12.txt
echo "A third file for testing.
The word TEST is here.
This file also has the keyword linux.
And some other random text." > file3.txt
echo "This is a different file.
It does not contain the word test.
This file also contains the keyword Linux in lowercase.
Another line for good measure." > file2.txt
echo "This is a test file.
It contains the word test multiple times.
This file also contains the keyword Linux.
And a final test line." > file1.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$

```

18. chmod (make a file as read only)

```

sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l file12.txt
-rwxrwxrwx 1 sunbeam sunbeam 449 Oct 25 17:54 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ chmod 0444 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l file12.txt
-r--r--r-- 1 sunbeam sunbeam 449 Oct 25 17:54 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$

```

19. chmod (make a file as read write and excute)

```

sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l file12.txt
-r--r--r-- 1 sunbeam sunbeam 449 Oct 25 17:54 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ chmod 0777 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l file12.txt
-rwxrwxrwx 1 sunbeam sunbeam 449 Oct 25 17:54 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$

```


20. wc (using redirection) {Take input from the file, count words and store output to other file}

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ wc file12.txt >> file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ file12.txt
file12.txt: command not found
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ cat file12.txt
 3  9 31 file14.txt
 1  4 20 file12.txt
 1  4 20 total
 3 12 55 file12.txt
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

21. who | wc (using pipe) {Count words in output of “who” command}

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ who | wc
      5      46
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

22. head, tail (using pipe) {Print lines 5 to 10 from the given file}.

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ head file12.txt | tail file12.txt
Extracts fields 1 using comma (",") as a delimiter.
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f2
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f5
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f1,2
Extracts fields 1 and 2, using comma (",") as a delimiter.
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f2,3
cut -d ',' -f1,3 file.csv
Extracts fields 1 and 3, using comma (",") as a delimiter.
ls -l -i -s | cut -d " " -f2
Extracts the second field (file size) from the output of ls -l -i -s command.
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

23. sort , uniq (using pipe) {Print the uniq values from the unsorted file}.

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ sort file12.txt | uniq file12.txt
 3  9 31 file14.txt
 1  4 20 file12.txt
 1  4 20 total
 3 12 55 file12.txt
Using pipe, we can redirect output of any command to the input of any other command.
Two processes are connected using pipe operator (|).
Two processes run simultaneously and are automatically rescheduled as data flows between them.
If you don't use pipes, you must use several steps to do single task.
: Specify a delimiter (default is tab)
-f : Specify fields (columns) to extract
-c : Specify character positions to extract
Examples:
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f1
Extracts fields 1 using comma (",") as a delimiter.
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f2
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f5
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f1,2
Extracts fields 1 and 2, using comma (",") as a delimiter.
echo "DBDA DESD DAC DMC DITISS" | cut -d " " -f2,3
cut -d ',' -f1,3 file.csv
Extracts fields 1 and 3, using comma (",") as a delimiter.
ls -l -i -s | cut -d " " -f2
Extracts the second field (file size) from the output of ls -l -i -s command.
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

24. who, grep (using pipe) { Display all users not using tty2, tty3, tty4}.

```
grep: ,tty3: no such file or directory
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ who | grep 'tty2'
sunbeam  tty2          2025-10-25 15:00 (tty2)
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

25. ls { Display all files in ascending order of their file size }.

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls | sort -s
12:05
44.htm
44.txt
cat
Desktop
Documents
Downloads
file12.txt
file13.txt
file14.txt
file15.txt
file16.txt
file17.txt
file18.txt
file19.txt
file20.txt
file3
five
four
Music
myName
myNewfile
one
Pictures
prac folder
Public
sdm
Shivam
snap
sunbeam
sunbeam_submission
Templates
three
two
Videos
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ sort -s
```


26. ls, grep (using pipe) { Count number of files and directories from current directory}.

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l | wc -l
36
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l | grep "^d" | wc -l
17
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l | grep "^-" | wc -l
17
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

27. ls, grep (using pipe) { Display only hidden files in a directory }.

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ ls -l | grep "\." | wc -l
11
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

28. tr { Convert file contents into uppercase , also learn "-s" option of this command }.

```
11
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "Shivam" | tr 'a-z' 'A-Z'
SHIVAM
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "SHivam" | tr 'a-z' 'A-Z'
SHIVAM
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "Shivam" | tr 'A-Z' 'a-z'
shivam
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "hello woorld" | tr -s 'o'
hello world
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```

29. cut { Split the file contents using delimiter and print required fields }

```
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "first second third fourth fifth sixth" | cut -d " " -f1
first
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "first second third fourth fifth sixth" | cut -d " " -f2,2,3,4
second third fourth
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "first second third fourth fifth sixth" | cut -c " " -f2,2,3,4
cut: only one type of list may be specified
Try 'cut --help' for more information.
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$ echo "first,second,third,fourth,fifth,sixth" | cut -d "," -f1,1,2,3,4
first,second,third,fourth
sunbeam@sunbeam-Lenovo-Ideapad-V15-ADA:~$
```